

- rozkład wykładniczy:

$$\rho = e^{-|\vec{r}|/r_0} \cdot \frac{1}{8\pi r_0^3}$$

chwilkowo opuszczamy $\hbar/2\pi$

$$\int e^{-|\vec{r}|/r_0} e^{i\vec{q}\cdot\vec{r}} d^3r = \int_0^\infty \int_0^\pi \int_0^{2\pi} e^{-|\vec{r}|/r_0} e^{iq\cdot r \cos(\theta)} r^2 \sin(\theta) dr d\theta d\phi =$$

wybieramy oś Z zgodnie z $\vec{q}$

$$4\pi \int_0^\infty e^{-|\vec{r}|/r_0} \frac{\sin(q\cdot r)}{q\cdot r} r^2 dr = \frac{4\pi}{q} \int_0^\infty e^{-|\vec{r}|/r_0} \sin(q\cdot r) r dr =$$

$$\frac{8\pi}{r_0} \frac{1}{[q^2 + 1/r_0^2]^2} = \frac{8\pi r_0^3}{[q^2 r_0^2 + 1]^2}$$

$$F(\vec{q}^2) = \frac{1}{8\pi r_0^3} \int e^{-|\vec{r}|/r_0} e^{i\vec{q}\cdot\vec{r}/\hbar} d^3r = \frac{1}{[q^2 r_0^2 / \hbar^2 + 1]^2}$$

$\int_0^\infty x e^{-ax} \sin bx dx = \frac{2ab}{(a^2 + b^2)^2} \quad (a > 0)$

**formuła
dipolowa**